CSCI 4521 - YASHWARDHAN PANCHAL
ROAD SIGN RECOGNITION WITH GTSRB

Objective: Classification of traffic/road signs images.
Dataset: German Traffic Sign Recognition Benchmark - 43 Classes - 40K Images

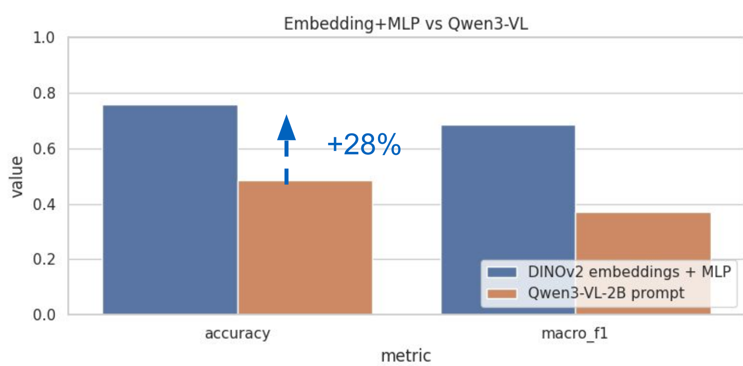
PIPELINE DIAGRAM

→ **Frozen DINOv2 Backbone** (1024x1) → **Custom Pytorch MLP** (1024 -> 512 -> 128 -> 43) → **Class Prediction**

LLM PROMPT

"You are classifying a GTSRB image. Choose exactly one label from: [List of 43 Classes]. Return only the label word."

KEY RESULTS



DINOv2 + MLP: 76.6% accuracy (Full test set)
Qwen3-VL Baseline: 48.5% accuracy (200-sample subset)
The Takeaway: Specialized training outperformed the zero-shot foundation model by **+28%**. VLMs struggle with fine-grained classification on low-resolution images compared to dedicated feature extractors.

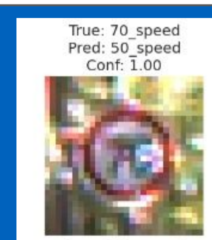
VISUAL ANALYSIS

Issue: Qwen3-VL confused "Turn Right" with "Attention Left Turn"

Cause: Foundation models often struggle with strict geometric and directional reasoning without few-shot examples to anchor them.

Issue: The MLP frequently confused 70km/h with 50km/h.

Cause: At low resolution, the global DINOv2 embedding prioritizes the dominant red circle over the blurry center text.



CONCLUSION

Specialization Wins: Dedicated feature extraction is vital for low-resolution road signs; general VLMs currently lack the fine-grained precision needed.
Engineering Trade-off: The MLP offers **sub-millisecond latency** and deterministic reliability, whereas the VLM is slower and prone to parsing errors.
Final Verdict: Deploy **DINOv2 + MLP** for real-time safety. Use **VLMs** for offline data labeling and edge-case analysis.